AlignAI: Personalized Physiotherapy System for Stroke Patients

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1 Introduction

Stroke is one of the leading causes of disability worldwide, affecting over **12.2 million people annually** and leaving survivors with significant physical and cognitive challenges. Despite advancements in medical care, rehabilitation often requires continuous, personalized support—something traditional systems struggle to deliver. Enter the **AlignAI,** an AI Physiotherapist for Stroke Patients Rehabilitation, a groundbreaking system that leverages the power of cutting-edge AI to revolutionize post-stroke recovery.

1.1 Purpose of the System

The primary objective of this project is to bridge the gap between stroke patients and effective rehabilitation by providing:

- **Personalized Assistance**: A conversational AI that answers queries related to stroke recovery, offering immediate and reliable guidance.
- **Live Progress Monitoring**: Real-time tracking of rehabilitation exercises using advanced computer vision techniques, ensuring patients adhere to their therapy plans.
- **Access to Latest Knowledge**: An intelligent Retrieval-Augmented Generation (RAG) system that keeps patients and caregivers updated with the latest medical literature and precautions.

1.2 Target Audience

The system is designed to benefit:

- **Stroke Patients**: Offering a virtual rehabilitation assistant to support their recovery journey.
- **Families and Caregivers**: Equipping them with tools and information to assist patients more effectively.
- **Rehabilitation Professionals**: Providing insights and tools to enhance patient monitoring and therapy customization.

By combining conversational AI, computer vision, and advanced data retrieval, this system aims to redefine rehabilitation, making it more accessible, efficient, and personalized.

From an academic perspective, this project represents a significant intersection of AI and healthcare. It offers a practical application of emerging technologies, demonstrating their potential to enhance patient care. The project also reflects a growing interest in leveraging technology to address complex health challenges, aligning with current trends in research and development. The objectives of AlignAI are clear and ambitious: to develop a system that provides personalized rehabilitation support, tracks patient progress with high accuracy, and adapts recommendations based on individual performance. The deliverables include a functional chatbot, a computer vision system for exercise monitoring, and a robust database for storing and analyzing patient data. While the project has great potential, it is important to acknowledge its limitations. The current implementation focuses on specific aspects of rehabilitation and may not encompass all possible needs of stroke patients. However, the foundation laid by AlignAI sets the stage for future advancements and improvements in the field.

2 System Overview

The **AlignAI system** is designed with a modular architecture to ensure seamless integration of its components and efficient operation. At its core, the system combines **large language model (LLM), computer vision**, and **data management** technologies to deliver a comprehensive physiotherapy solution. Figure 1 shows the workflow diagram of our project, it's understanding will be beneficial for the reader to understand our project.

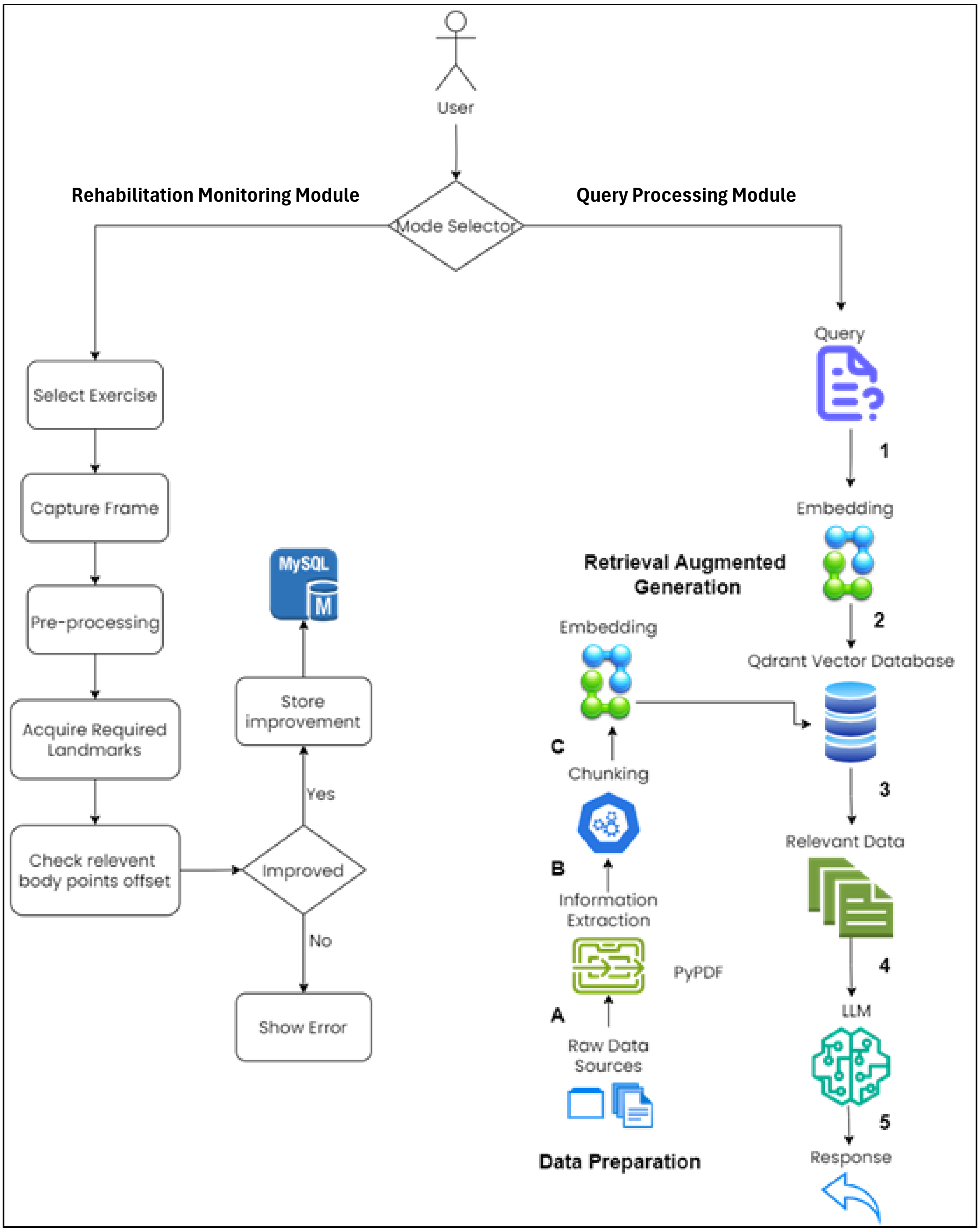

Figure 1: Workflow Diagram of AlignAI

The architecture consists of three main modules:

1. The **Query Processing Module**, powered by the LLaMA 3.1 large language model, handles user queries related to stroke rehabilitation. It interprets natural language inputs and generates accurate, context-aware responses.
2. The **Rehabilitation Monitoring Module**, leveraging Mediapipe and OpenCV, uses camera input to monitor a patient's exercise routines in real-time. It detects body landmarks and analyzes movements to assess progress and provide feedback.
3. The **Knowledge Retrieval Module**, which incorporates a Qdrant vector database in a **Retrieval-Augmented Generation (RAG)** framework, stores and retrieves the latest rehabilitation resources and precautionary information. This module ensures users receive up-to-date, reliable guidance.

User data, including progress reports and interaction history, is securely managed in MongoDB, ensuring both privacy and accessibility. These components work together in a pipeline to provide a seamless user experience, with minimal latency and high accuracy.

2.1 Technologies Used

The development of AlignAI relies on a suite of state-of-the-art technologies. At the heart of the NLP functionality is **LLaMA 3.1**, a high-performance large language model that interprets and generates natural language responses tailored to user queries. The **Qdrant vector database** plays a critical role in storing and retrieving data encoded as **384-dimensional vectors** by the **all-MiniLM-L12-v2** sentence transformer, enabling efficient handling of large-scale text information.

For real-time rehabilitation monitoring, **Mediapipe** and **OpenCV** provide robust computer vision capabilities, enabling precise detection and tracking of body movements. Finally, **MongoDB** ensures secure and efficient storage of user data, acting as a reliable backend for managing patient information.

2.2 Tools Used

In developing the AlignAI project, several tools and technologies were employed to build a comprehensive and effective rehabilitation system. Each tool was chosen for its specific capabilities and role in achieving the project’s goals.

2.2.1 Gradio

Gradio is a user-friendly tool used to create interactive web interfaces for machine learning models. In AlignAI, Gradio is used to design and develop the user interface for the chatbot. It allows us to build a straightforward and accessible interface where users can interact with the chatbot, receive exercise recommendations, and track their progress. Gradio’s simplicity and flexibility make it an ideal choice for integrating the AI model with user interactions.

2.2.2 PyPDF2

To extract text from the PDF versions of the rehabilitation guides, we used PyPDF2. This Python library helps in reading and processing PDF files, allowing us to convert the text into a format that can be analyzed and used for generating recommendations. PyPDF2 handles the extraction of raw text efficiently, which is crucial for the subsequent steps of text processing.

2.2.3 SentenceTransformer

The SentenceTransformer model is employed to generate embeddings for the text extracted from the rehabilitation guides. Using the 'all-MiniLM-L12-v2' model, we convert chunks of text into numerical vectors that capture the semantic meaning of the content. These embeddings are essential for understanding and retrieving relevant information based on user queries.

2.2.4 Qdrant

Qdrant is a vector search engine used to store and manage the text embeddings. It provides a scalable and efficient way to perform similarity searches, helping us find the most relevant chunks of text in response to user queries. By using Qdrant, we ensure that the chatbot can quickly access and retrieve pertinent information, enhancing the quality and accuracy of the recommendations provided.

2.2.5 Mediapipe

For monitoring and analyzing the patient's exercise performance, we used Mediapipe. This tool allows us to track keypoints on the body during exercises, providing real-time feedback on the patient's movements. Mediapipe's capabilities in body pose estimation are instrumental in assessing the effectiveness of the exercises and tracking improvements over time.

2.2.6 MongoDB

MongoDB is used as the database management system for storing personalized information about the patients and their progress. It provides a reliable and structured way to keep track of data, including personal details and exercise performance metrics. MongoDB's relational database features enable efficient querying and updating of patient records, which is vital for monitoring progress and providing tailored recommendations.

These tools collectively enable AlignAI to offer a robust and effective solution for stroke rehabilitation. Each tool was selected based on its ability to address specific needs within the project, ensuring a seamless integration of various components and functionalities.

2.3 System Features

The AlignAI system offers a range of features aimed at improving stroke rehabilitation. One of its standout capabilities is **query answering**, where users can interact with the system to ask questions about rehabilitation exercises, best practices, and recovery timelines. The system provides detailed and accurate responses, ensuring users have access to essential information when they need it most.

Another critical feature is **live rehabilitation monitoring**, which uses camera input to track patient movements during exercises. This functionality not only provides real-time feedback but also ensures patients perform their routines correctly, helping to prevent potential injuries and maximize recovery outcomes.

Lastly, the system incorporates a **resource retrieval feature** powered by the RAG setup, allowing it to provide users with the latest medical literature, guidelines, and precautionary information about stroke rehabilitation. This ensures patients and caregivers are always informed about the most effective and safe recovery practices.

By integrating these advanced technologies and features, AlignAI delivers a powerful, user-friendly platform to support stroke patients on their path to recovery.

3 Methodology

The methodology of the AlignAI project outlines the step-by-step approach used to develop and implement the system, which includes a chatbot for recommendations, computer vision for exercise monitoring, and a personalized database. This section details how each component was designed and integrated to achieve the project's objectives. Figure 2 illustrates the workflow diagram of AlignAI.

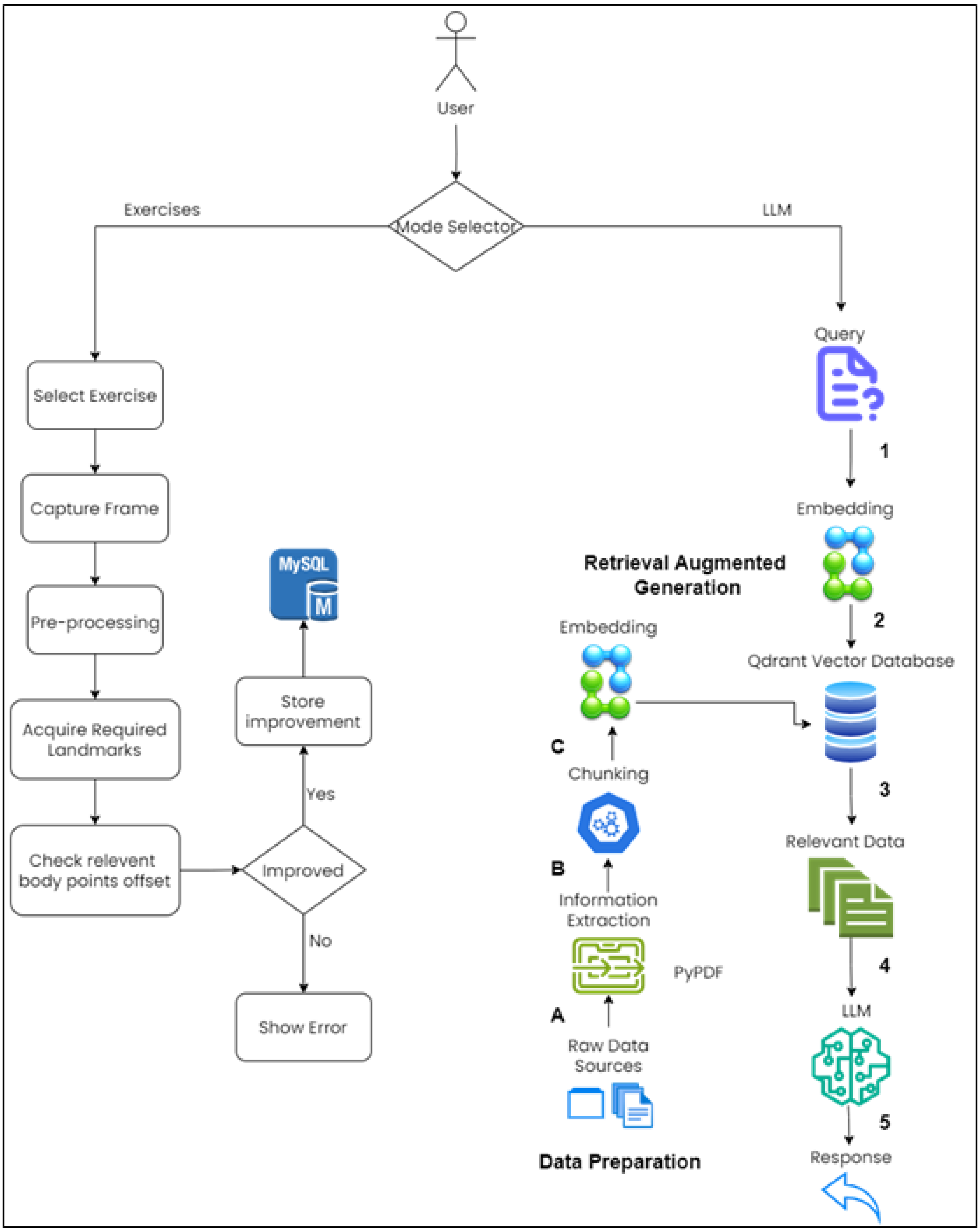

Figure 2: Workflow diagram of AlignAI

3.1 Text Extraction

To build a knowledge base for the chatbot, we needed to extract relevant information from the chosen reference books: **ASA Hope Stroke Recovery Guide** and **O'Sullivan & Schmitz's**

Physical Rehabilitation. This process began with text extraction from PDF files. We used the PyPDF2 library to read and extract text from the PDFs. Then preprocessing techniques were applied. This included removing unwanted characters such as double spaces and zero-width spaces to ensure the text was clean and usable for further processing.

3.2 Chunking

Given the volume of text extracted, we needed to divide it into manageable sections. Two chunking techniques were employed to handle this effectively:

- *Semantic Chunking:* This technique involves grouping sentences that are similar in meaning. We used BERT embeddings to represent each sentence as a numerical vector. Sentences were clustered into groups using KMeans, which helped in creating coherent chunks. Each chunk was constrained by a maximum size to ensure it remained relevant and understandable.

- *Recursive Chunking*: For text that was too lengthy, we applied recursive chunking. Long sentences were split into smaller chunks by breaking them down into shorter segments until they fit within the size limits. This ensured that all text chunks were manageable and could be processed efficiently.

3.3 Embeddings

With the text divided into chunks, we generated embeddings to represent these chunks numerically. We used the SentenceTransformer model (`all-MiniLM-L12-v2`) to convert the text chunks into vectors. These embeddings capture the semantic meaning of the text and are crucial for the next step of our methodology.

3.4 Vector Database

To store and retrieve the text chunks and their embeddings, we used Qdrant, a vector search engine. We prepared the text chunks and their embeddings, encapsulated them into PointStruct objects, and inserted them into a Qdrant collection. The Qdrant collection was set up with a dimensionality of 384, matching the SentenceTransformer embeddings. We used the cosine distance metric to measure similarity between embeddings, allowing us to efficiently perform similarity searches and retrieve relevant information.

3.5 LLM

When a user query, such as "Suggest exercises for stroke patients," is received, it is embedded using the SentenceTransformer model. This query embedding is then used to perform a similarity search in Qdrant, retrieving the most relevant text chunks from the stored data. The retrieved chunks are sent to the LLaMA 3.1 405B instruct model for natural language generation. We created a custom prompt that incorporates the retrieved chunks, providing context to the model. The LLaMA model then generates a response based on this context, offering tailored exercise recommendations.

3.6 Computer Vision

For the computer vision component, we utilized Mediapipe to monitor exercise performance. Mediapipe tracks keypoints on the body during exercises, allowing us to assess the accuracy and effectiveness of each exercise. This real-time feedback is crucial for understanding patient progress and making necessary adjustments to their rehabilitation routine.

3.7 Personalized Data Storage

We used MongoDB to store and manage personalized data. The database tracks user information, exercise performance, and improvements over time. This data is essential for generating detailed reports and visualizations in the dashboard.

4 Technical Specifications

4.1 Integration Workflow

The **AlignAI: AI Physiotherapist for Stroke Patients** operates through a meticulously designed workflow that ensures seamless integration of its components. When a user interacts with the system, their query or input is first directed to the **Query Processing Module**, where the LLaMA 3.1 large language model interprets the input. If the input is a query requiring contextual information, the system consults the **Knowledge Retrieval Module**, which uses the Qdrant vector database to fetch relevant information.

Simultaneously, if the user engages with the **Rehabilitation Monitoring Module**, a live feed from the camera is processed by **Mediapipe** and **OpenCV**. These tools identify key body landmarks and track movements to analyze exercise performance. The results are then compared with predefined standards to provide real-time feedback.

The **Knowledge Retrieval Module** works hand-in-hand with the **Embedding Processor**, which converts textual data into vectorized representations using the **all-MiniLM-L12-v2 sentence transformer**. This process ensures rapid and accurate retrieval of the most relevant information from the Qdrant database.

All interactions, user data, and progress metrics are stored in **MongoDB**, enabling secure and organized data management. The integration across modules is facilitated by APIs and optimized pipelines, ensuring minimal latency and a smooth user experience.

4.2 Embedding Process

The **sentence-to-vector transformation** is a pivotal aspect of the AlignAI system, enabling it to handle large-scale textual data with high efficiency. This process is powered by the **all-MiniLM-L12-v2 sentence transformer**, a state-of-the-art model designed to convert natural language sentences and paragraphs into dense vector representations within a **384-dimensional space**.

When a new resource, such as a rehabilitation guide or precautionary advice, is added to the system, the text is first tokenized and processed by the embedding model. The model extracts semantic information and encodes it into a fixed-length vector that captures the meaning and context of the input text.

These vectors are then stored in the **Qdrant vector database**, which organizes and indexes them for efficient retrieval. During a query, the system generates a vector representation of the input using the same embedding model and compares it against the stored vectors using similarity measures like cosine similarity. The closest match is retrieved and provided as a response, ensuring the information is both relevant and accurate.

This embedding process, combined with the Retrieval-Augmented Generation (RAG) framework, enables AlignAI to deliver precise, contextually rich responses to user queries, enhancing the overall rehabilitation experience.

5 Setup and Deployment

This section will guide the reader on how to install and run the project on their system, including system requirements, installation steps, and instructions for starting and using the system.

5.1 System Requirements:

5.1.1 Hardware Requirements

To ensure smooth operation of **AlignAI: AI Physiotherapist for Stroke Patients**, the following minimum hardware specifications are recommended:

- **CPU**: Multi-core processor with at least 2.5 GHz (Intel i5/i7 or equivalent).
- **RAM**: 8 GB or higher for efficient processing of models and data.
- **Storage**: Minimum of 1 GB free disk space for the models, databases, and application files.
- **Camera**: A high-resolution camera (720p or higher) for live rehabilitation monitoring.

5.1.2 Software Requirements

The system is designed to run on the following software platforms:

- **Operating System**: Windows 10/11, macOS 10.15 or later, or a Linux distribution (Ubuntu 20.04 or later recommended).
- **Python**: Version 3.9 or later.
- **Database**: MongoDB (v5.0 or later) and Qdrant (v1.0 or later).

5.1.3 Dependencies

To ensure smooth operation of AlignAI, the project relies on several Python libraries and frameworks. These dependencies are divided into two modules: **Chatbot** and **Backend**. Each module has its own requirements.txt file for simplified installation.

1. **Chatbot Module Dependencies**: The chatbot functionality uses libraries such as Gradio for the user interface, Mediapipe and OpenCV for exercise monitoring, and Qdrant for managing embeddings. It also requires sentence transformers for embedding generation and PyMongo for database interaction.
2. **Backend Module Dependencies**: The backend module uses Flask for API development, Flask-CORS for handling cross-origin requests, PyMongo for MongoDB operations, and Werkzeug for security-related functions like password hashing.

Installation of Dependencies

Download this github repo. The dependencies for each module are listed in the respective requirements.txt files located in the project directory. Follow these steps to install the required libraries:

1. Navigate to the **Chatbot** directory and run:

```
pip install -r requirements.txt
```

2. Navigate to the **Backend** directory and run:

```
pip install -r requirements.txt
```

These commands will automatically install all the necessary libraries and their dependencies, ensuring the system is ready for deployment.

5.2 Databases Setup

The AlignAI requires two primary databases: **Qdrant** for vector storage and **MongoDB Atlas** for user data and system records. Below are the detailed setup instructions for each.

1. Qdrant: Vector Database

Step 1: Register for a Qdrant Cloud Account

1. Go to the Qdrant Cloud website and sign up using your email, Google, or GitHub credentials.
2. After registering, log in to your account to access the dashboard.

Step 2: Create a Qdrant Cluster

1. Navigate to the **Overview** section and follow the instructions under **Create First Cluster**.
2. Choose a name for your cluster and initiate the creation process.

Step 3: Retrieve API Key

1. Once your cluster is created, an **API key** will be generated.
2. Save this API key, as it will be required to authenticate your application with the Qdrant cluster.

Step 4: Access the Cluster Dashboard

1. Go to the **Clusters** section and locate your newly created cluster.
2. Under **Actions**, open the **Dashboard** for your cluster.
3. Paste your API key into the dashboard to authenticate.

Step 5: Integrate Qdrant with the Application

Use the **Qdrant SDK** to connect your application to the Qdrant cluster. Below is an example Python snippet to initialize the connection:

```
from qdrant_client import QdrantClient

qdrant_client = QdrantClient(
    host="your-cluster-url",  # e.g., "xyz-example.eu-
central.aws.cloud.qdrant.io"
    api_key="your-api-key"   # Replace with your actual API key
)
```

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2. MongoDB Atlas: NoSQL Database

Step 1: Set Up a MongoDB Atlas Account

3. Go to the [MongoDB Atlas website] and create an account.
4. After signing up, log in to the Atlas dashboard.

Step 2: Create a Cluster

1. Click on **Create Cluster** in the dashboard.
2. Choose a cloud provider (AWS, Azure, or GCP) and a region close to your location for optimal performance.
3. Select the cluster tier based on your project's requirements (the free tier is suitable for development).

Step 3: Configure Database Access

1. Under the **Database Access** tab, create a new database user and set a password.
2. Assign appropriate permissions to the user, such as read/write access.

Step 4: Whitelist Your IP Address

1. Navigate to the **Network Access** tab and click **Add IP Address**.
2. Add your IP or choose **Allow Access from Anywhere** for development purposes.

Step 5: Connect to MongoDB Atlas

1. Go to the **Clusters** section and click **Connect**.
2. Choose the connection method as **Connect your application**.
3. Copy the connection string, which will look like this:

   ```
   mongodb+srv://<username>:<password>@cluster0.mongodb.net/?retryWrites
   =true&w=majority
   ```

Step 6: Integrate MongoDB with the Application Use the connection string to connect your application to MongoDB. Below is an example Python snippet using `pymongo`:

```
from pymongo import MongoClient

client = MongoClient("your-connection-string")  # Replace with your actual
connection string
db = client["database_name"]  # Replace with your desired database name
```

5.3 API Keys

The AlignAI requires an API key to access the **LLaMA 3.1 405b** model. Below are the steps to obtain and configure the API key for your project.

5.3.1 *Obtaining the LLaMA 3.1 405b API Key*

1. **Visit NVIDIA Build Platform**:
   Go to the official NVIDIA Build website.
2. **Create an Account**:
   - Sign up for an account using your email or log in if you already have one.
   - Upon creating a new account, you will receive **1,000 free credits** to get started.
3. **Search for LLaMA 3.1 405b**:
   - Use the search bar to find the **LLaMA 3.1 405b** model.
   - Click on the model to view its details.
4. **Retrieve the API Key**:
   - Scroll down the model page and click on the **Get API Key** button.
   - A unique API key will be generated for you.
   - Copy this API key, as you will need it for project configuration.

5.3.2 *Configuring the API Key*

1. **Locate the Configuration File**:
   - In your project directory, find the file named `config.json`.
2. **Paste the API Key**:
   - Open the `config.json` file in a text editor.
   - Add the API key under the appropriate key-value pair.
3. **Save the File**:
   - Ensure the changes are saved before proceeding.

5.4 Running the Project

To start the **AlignAI: AI Physiotherapist for Stroke Patients**, follow the steps below to launch the chatbot, backend, and user interface components:

1. **Run the Chatbot Module**:
   - Navigate to the `chatbot` folder in your project directory.
   - Execute the following command to launch the chatbot interface using Gradio:

     ```
     gradio app.py
     ```

   - This will start the chatbot application, allowing users to interact with the physiotherapy system.
2. **Run the Backend Module**:
   - Navigate to the `backend` folder in your project directory.
   - Start the backend server using Flask with the following command:

     ```
     flask run
     ```

   - The backend handles API requests and database interactions.

3. **Launch the User Interface**:
   - Open the `landing_page` folder in your project directory.
   - Locate the `index.html` file and open it in your web browser.
4. **Hurray! The Project is Running**
   - With the chatbot, backend, and user interface running, you can now fully utilize the **AlignAI** system.
   - Test the features, interact with the chatbot, and explore the rehabilitation monitoring functionalities.

6 Comparison of Results with Existing Models

In assessing the effectiveness of AlignAI, it's important to compare its results with those of existing models in the field of stroke rehabilitation and personalized physiotherapy. This comparison helps highlight the strengths and areas for improvement of AlignAI. Current models and systems in stroke rehabilitation often focus on either traditional methods or technology-driven solutions. Many of the traditional approaches rely heavily on manual tracking and standardized exercise routines, which can lack personalization. Some newer systems incorporate technology, such as:

6.1 Virtual Reality (VR) Rehabilitation Systems:

These systems use VR to create immersive rehabilitation experiences. They offer interactive exercises that can adapt to patient progress but often require specialized equipment and are limited by their inability to provide real-time feedback on exercise execution.

6.2 Wearable Devices:

Wearable sensors track physical activity and movement, providing data on exercise performance. While useful, these devices may not offer detailed feedback on specific exercise techniques or personalize recommendations based on individual progress.

6.3 Telehealth Platforms:

These platforms facilitate remote physiotherapy sessions and may include features for tracking exercise adherence and progress. However, they generally depend on manual input from patients and therapists, which can introduce variability and limit the ability to provide real-time, personalized feedback.

6.4 AlignAI’s Performance Compared to Existing Models

6.4.1 Personalization and Feedback

AlignAI stands out for its personalized approach to stroke rehabilitation. The integrated chatbot provides tailored exercise recommendations based on individual needs, which is a step beyond the generalized routines offered by many existing systems. By using a database that tracks each patient’s progress, AlignAI can adjust recommendations in real time, something that traditional and even some modern systems may not achieve as effectively.

6.4.2 Exercise Monitoring

The computer vision component of AlignAI, which uses Mediapipe to monitor and analyze exercise performance, offers detailed feedback on exercise execution. This feature provides a level of precision and real-time assessment that goes beyond what is typically available with wearable devices or VR systems. Existing models often lack this depth of analysis, relying instead on less detailed or manual forms of feedback.

6.4.3 Progress Tracking and Adaptation

AlignAI’s use of MongoDB for personalized data storage allows for a comprehensive tracking of patient progress over time. This feature enables the system to highlight improvements, track exercise completion rates, and suggest modifications based on detailed historical data. While some existing models offer progress tracking, AlignAI's approach is more nuanced, providing a clearer picture of patient progress and adapting recommendations based on this data.

6.4.4 *Ease of Use and Accessibility*

AlignAI's user interface, powered by Gradio, is designed to be intuitive and accessible, making it easier for patients to interact with the system. Compared to the often-complex setups of VR systems or the more manual interaction required by telehealth platforms, AlignAI aims to provide a more user-friendly experience.

7 Conclusion

AlignAI represents a significant step forward in stroke rehabilitation, integrating modern technology to address key challenges faced by patients and healthcare providers. Through its three main components—an AI-driven chatbot, a computer vision system, and a personalized database—AlignAI aims to offer a more tailored and effective rehabilitation experience.

The chatbot provides personalized exercise recommendations based on a patient's needs, while the computer vision system monitors exercise performance and tracks improvements. The MongoDB database supports this process by storing and analyzing patient data, ensuring that progress can be accurately measured and adjusted over time.

The project's approach not only enhances the rehabilitation process but also makes it more adaptable to individual patients. By providing detailed feedback and tracking improvements, AlignAI helps patients stay engaged and motivated throughout their recovery journey. This personalized approach is expected to lead to better outcomes and a more efficient rehabilitation process.

While the current implementation focuses on specific aspects of stroke rehabilitation, it lays a solid foundation for future developments. There is room for expanding the system to address additional needs and incorporate more advanced features. Overall, AlignAI demonstrates the potential of combining AI and technology to improve patient care and support in the field of rehabilitation.

In conclusion, AlignAI not only contributes to advancing stroke rehabilitation but also sets a precedent for integrating technology into healthcare. It represents a promising step towards more personalized, effective, and engaging rehabilitation practices.